

## FREMONT, NE, USA

WMO: 725564

Lat: 41.449N

Lon: 96.520W

Elev: 367

StdP: 97.00

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 04924

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -19.4      | -17.1 | -23.2                         | 0.5 | -18.3 | -21.1 | 0.6 | -16.0 | 14.5                  | -4.6 | 12.7 | -3.7 | 3.1                   | 290  | 0.545 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.5                   | 34.1            | 23.6 | 32.7 | 23.1 | 31.3 | 22.5 | 25.7                | 31.2 | 24.8 | 30.2 | 23.9 | 29.2 | 4.8                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.0                            | 19.8 | 29.4 | 22.9 | 18.4 | 28.2 | 22.3 | 17.7 | 27.6 | 81.6          | 31.1 | 77.0 | 30.3 | 73.4 | 28.9 | 28.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 11.8          | 10.3 | 8.8 | -24.3                      | 36.6 | 2.1                | 1.5 | -25.7                                              | 37.7 | -26.9      | 38.6 | -28.1      | 39.4 | -29.6      | 40.5 |    |    |
| (5)               |      |     | -24.6                      | 27.1 | 1.9                | 0.9 | -26.0                                              | 27.7 | -27.1      | 28.2 | -28.2      | 28.7 | -29.6      | 29.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.7   | -4.4  | -2.4  | 4.1   | 10.7  | 16.7  | 22.3  | 24.8  | 23.0  | 19.2  | 11.9  | 4.2   | -2.6  |  |  |
|                                                                        | DBStd       | 11.56  | 6.66  | 6.56  | 6.75  | 5.53  | 4.94  | 3.82  | 3.39  | 3.16  | 4.69  | 5.33  | 5.71  | 6.33  |  |  |
|                                                                        | HDD10.0     | 1682   | 445   | 349   | 204   | 58    | 6     | 0     | 0     | 0     | 1     | 41    | 188   | 391   |  |  |
|                                                                        | HDD18.3     | 3396   | 703   | 582   | 441   | 236   | 91    | 10    | 2     | 3     | 46    | 210   | 424   | 649   |  |  |
|                                                                        | CDD10.0     | 1941   | 0     | 1     | 23    | 77    | 214   | 370   | 459   | 404   | 278   | 99    | 13    | 1     |  |  |
|                                                                        | CDD18.3     | 611    | 0     | 0     | 1     | 6     | 41    | 130   | 202   | 149   | 73    | 9     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 6072   | 0     | 0     | 17    | 89    | 424   | 1293  | 2112  | 1321  | 708   | 105   | 3     | 0     |  |  |
|                                                                        | CDH26.7     | 2282   | 0     | 0     | 3     | 22    | 136   | 486   | 891   | 472   | 248   | 26    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.8    | 3.9   | 4.0   | 4.4   | 4.7   | 4.2   | 3.8   | 3.0   | 2.8   | 3.4   | 3.8   | 3.9   | 3.9   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 766    | 19    | 23    | 41    | 76    | 119   | 125   | 89    | 99    | 69    | 52    | 35    | 23    |  |  |
|                                                                        | PrecMax     | 1025   | 42    | 63    | 88    | 129   | 218   | 275   | 269   | 292   | 181   | 143   | 75    | 61    |  |  |
|                                                                        | PrecMin     | 514    | 4     | 3     | 0     | 13    | 34    | 33    | 18    | 12    | 16    | 3     | 1     | 0     |  |  |
|                                                                        | PrecStd     | 134    | 10    | 14    | 26    | 26    | 52    | 59    | 55    | 67    | 40    | 36    | 22    | 16    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.9  | 17.9  | 25.3  | 29.8  | 33.0  | 35.1  | 37.0  | 35.7  | 33.9  | 30.0  | 22.7  | 14.9  |  |  |
|                                                                        |             | MCWB   | 6.4   | 10.0  | 14.0  | 16.7  | 19.9  | 22.9  | 23.4  | 22.8  | 22.4  | 19.3  | 13.6  | 10.2  |  |  |
|                                                                        | 2%          | DB     | 9.9   | 12.8  | 21.8  | 26.0  | 30.1  | 32.8  | 34.9  | 32.8  | 32.0  | 26.2  | 19.0  | 11.3  |  |  |
|                                                                        |             | MCWB   | 5.3   | 7.6   | 13.2  | 15.2  | 19.4  | 22.4  | 24.2  | 24.0  | 21.6  | 16.7  | 12.0  | 6.8   |  |  |
|                                                                        | 5%          | DB     | 7.1   | 9.8   | 17.9  | 22.8  | 27.7  | 31.2  | 32.8  | 31.2  | 29.8  | 23.3  | 16.2  | 8.0   |  |  |
|                                                                        |             | MCWB   | 3.5   | 5.8   | 11.7  | 13.8  | 17.8  | 21.7  | 24.3  | 23.4  | 21.1  | 15.4  | 10.3  | 4.7   |  |  |
|                                                                        | 10%         | DB     | 4.0   | 6.8   | 14.9  | 20.2  | 25.6  | 29.7  | 31.3  | 29.3  | 27.5  | 21.1  | 13.0  | 5.9   |  |  |
|                                                                        |             | MCWB   | 1.7   | 3.7   | 9.5   | 12.6  | 17.2  | 20.9  | 23.6  | 22.6  | 19.7  | 14.3  | 8.3   | 3.0   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 7.4   | 12.1  | 17.1  | 18.4  | 22.4  | 25.4  | 27.3  | 26.7  | 24.4  | 20.9  | 15.4  | 11.4  |  |  |
|                                                                        |             | MCDB   | 12.6  | 14.7  | 21.8  | 26.1  | 28.7  | 31.1  | 32.7  | 31.4  | 29.8  | 26.7  | 19.3  | 13.7  |  |  |
|                                                                        | 2%          | WB     | 5.3   | 8.0   | 15.0  | 16.7  | 20.9  | 24.1  | 26.1  | 25.4  | 23.2  | 18.7  | 13.1  | 7.2   |  |  |
|                                                                        |             | MCDB   | 9.2   | 12.3  | 18.9  | 23.4  | 28.0  | 30.2  | 32.0  | 30.5  | 29.0  | 23.5  | 18.1  | 10.2  |  |  |
|                                                                        | 5%          | WB     | 3.7   | 5.7   | 12.2  | 15.1  | 19.7  | 23.1  | 25.2  | 24.4  | 22.3  | 16.9  | 10.9  | 5.0   |  |  |
|                                                                        |             | MCDB   | 6.6   | 9.5   | 17.9  | 21.2  | 25.8  | 29.3  | 30.9  | 29.3  | 27.8  | 21.5  | 15.4  | 8.4   |  |  |
|                                                                        | 10%         | WB     | 2.0   | 3.8   | 9.5   | 13.5  | 18.4  | 22.1  | 24.2  | 23.4  | 21.2  | 15.0  | 8.6   | 3.0   |  |  |
|                                                                        |             | MCDB   | 4.1   | 6.6   | 14.9  | 19.5  | 23.7  | 28.2  | 29.7  | 28.1  | 26.1  | 19.9  | 13.4  | 5.4   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.9   | 9.9   | 11.4  | 12.0  | 11.4  | 10.9  | 10.5  | 10.6  | 12.5  | 12.4  | 11.1  | 9.5   |  |  |
|                                                                        |             | MCDBR  | 13.8  | 14.1  | 16.7  | 16.3  | 14.3  | 12.3  | 11.7  | 12.5  | 14.0  | 15.6  | 15.6  | 13.2  |  |  |
|                                                                        | 5% WB       | MCWBR  | 9.6   | 9.3   | 9.4   | 8.2   | 6.4   | 5.3   | 5.1   | 5.5   | 5.9   | 7.8   | 9.2   | 9.3   |  |  |
|                                                                        |             | MCWBR  | 13.0  | 13.2  | 14.8  | 13.6  | 12.2  | 10.8  | 10.4  | 10.4  | 11.8  | 12.7  | 14.1  | 12.4  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.283 | 0.310 | 0.330 | 0.390 | 0.413 | 0.419 | 0.427 | 0.419 | 0.378 | 0.330 | 0.296 | 0.284 |  |  |
|                                                                        | taud        |        | 2.442 | 2.392 | 2.391 | 2.236 | 2.245 | 2.290 | 2.269 | 2.291 | 2.361 | 2.483 | 2.502 | 2.453 |  |  |
|                                                                        | Ebn at Noon |        | 878   | 900   | 919   | 879   | 865   | 857   | 846   | 842   | 856   | 866   | 857   | 843   |  |  |
|                                                                        | Edh at Noon |        | 77    | 94    | 106   | 134   | 136   | 131   | 132   | 125   | 109   | 86    | 72    | 70    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.00  | 2.94  | 3.81  | 4.92  | 5.80  | 6.48  | 6.53  | 5.60  | 4.60  | 3.23  | 2.26  | 1.68  |  |  |
|                                                                        | RadStd      |        | 0.19  | 0.17  | 0.34  | 0.35  | 0.43  | 0.48  | 0.41  | 0.39  | 0.29  | 0.37  | 0.15  | 0.17  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (39 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page